Cross-Domain Feature Augmentation for Domain Generalization

Yingnan Liu, Yingtian Zou, Rui Qiao, Fusheng Liu, Mong Li Lee, Wynne Hsu

National University of Singapore

IJCAI JEJU 2024

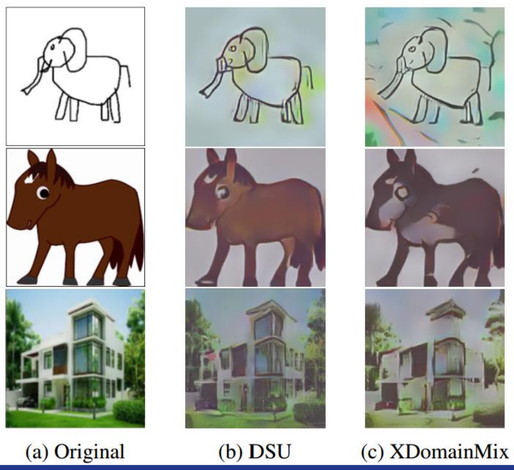
Introduction

Background

- Aim to learn a model that generalizes well to unseen **distributions** by learning **invariance** across training domains
- **Data augmentation** is widely used to learn invariance predictors
- Augmentation in **feature space** is more versatile than in input space
- **Feature semantics** are seldomly considered, which should be leveraged to improve augmentation **diversity**

Proposed XDomainMix

- View feature as a composition of **domain/class-specific/generic** components
- Perform **semantics-preserved** feature augmentation to increase augmentation **diversity** and learn **invariance**



(a) Original (b) DSU (c) XDomainMix

Methodology

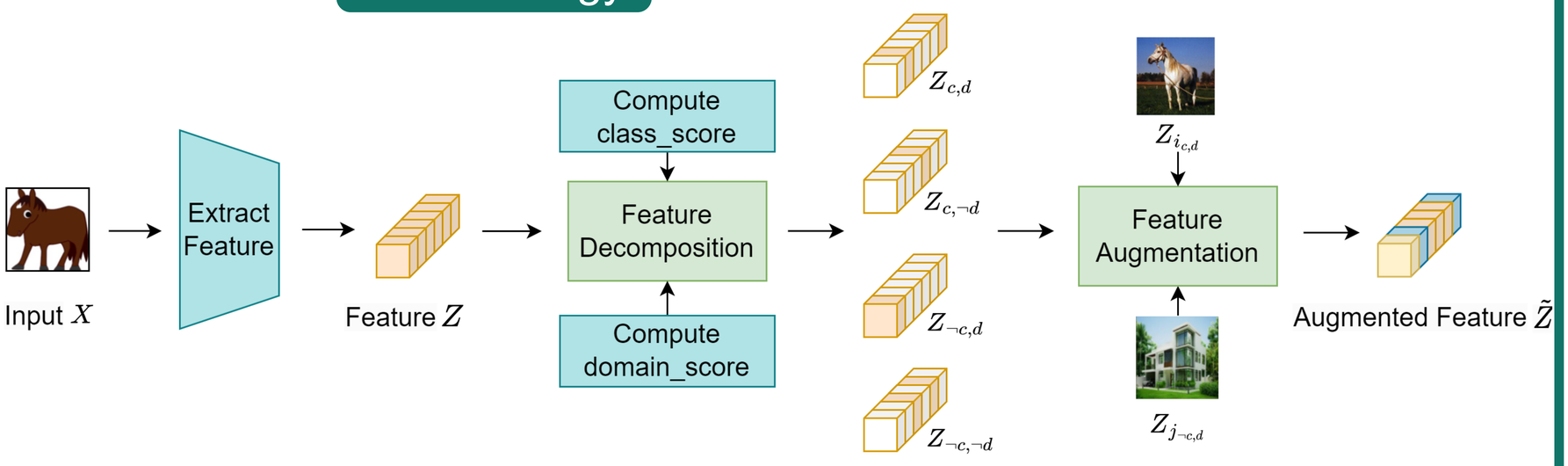
Feature Decomposition

Feature's semantics	Class-specific (Z_c)	Class-generic ($Z_{\neg c}$)
Domain-specific (Z_d)	$Z_{c,d}$	$Z_{\neg c,d}$
Domain-generic ($Z_{\neg d}$)	$Z_{c,\neg d}$	$Z_{\neg c,\neg d}$

Class/domain_score to calculate **contribution** to ground-truth class/domain prediction v:

$$\text{class/domain_score}(z) = \frac{\partial v}{\partial z} z$$

- Higher contribution, contain more class/domain-specific information



Feature Augmentation

To **reduce reliance** on domain-specific information, **augment the domain-specific feature component** to enhance diversity from a domain perspective:

$$\tilde{Z}_{c,d} = \lambda \cdot Z_{c,d} + (1-\lambda) \cdot Z_{i_{c,d}}$$

same class different domain sample i

$$\tilde{Z}_{\neg c,d} = \gamma \cdot Z_{\neg c,d} + (1-\gamma) \cdot Z_{j_{\neg c,d}}$$

different class different domain sample j

$$\tilde{Z} = \begin{cases} \tilde{Z}_{\neg c,d} + Z_{c,\neg d} + Z_{\neg c,\neg d} & \text{if } p \le p_{discard} \\ \tilde{Z}_{c,d} + \tilde{Z}_{\neg c,d} + Z_{c,\neg d} + Z_{\neg c,\neg d} & \text{otherwise} \end{cases}$$

To learn invariance and explore less activated features

$\tilde{Z}$ is assigned the **same class label as Z**

Performance Study

Domain Generalization

Method	Camelyon17	FMoW	PACS	TerraIncognita	DomainNet
ERM	70.3±6.4	32.3±1.3	85.5±0.2	46.1±1.8	43.8±0.1
GroupDRO	68.4±7.3	30.8±0.8	84.4±0.8	43.2±1.1	33.3±0.2
RSC	77.0±4.9^	32.6±0.5^	85.2±0.9	46.6±1.0	38.9±0.5
MixStyle	62.6±6.3^	32.9±0.5^	85.2±0.3	44.0±0.7	34.0±0.1
DSU	69.6±6.3^	32.5±0.6^	85.5±0.6^	41.5±0.9^	42.6±0.2^
LISA	77.1±6.5	35.5±0.7	83.1±0.2^	47.2±1.1^	42.3±0.3^
Fish	74.7±7.1	34.6±0.2	85.5±0.3	45.1±1.3	42.7±0.2
XDomainMix	**80.9±3.2**	**35.9±0.8**	**86.4±0.4**	**48.2±1.3**	**44.0±0.2**

Model Invariance

Representation invariance, measured by distance of covariance matrix of same class representations across domains:

Method	Camelyon17	FMoW	PACS	TerraIncognita	DomainNet
ERM	0.47±0.18	0.35±0.09	0.69±0.12	0.50±0.09	3.60±0.38
GroupDRO	0.21±0.02	0.40±0.04	0.77±0.16	0.43±0.05	2.13±0.09
RSC	0.32±0.17	0.55±0.14	42.3±12.8	29.9±3.02	17.3±1.43
MixStyle	0.28±0.26	0.36±0.05	0.69±0.03	0.38±0.09	3.39±0.17
DSU	**0.07±0.02**	0.32±0.02	0.33±0.04	9.18±1.84	4.61±0.24
LISA	0.19±0.05	0.29±0.05	0.04±0.00	0.13±0.01	0.72± 0.06
Fish	3.95±3.28	0.47±0.02	0.64±0.34	0.34±0.04	2.60±0.26
XDomainMix	0.19±0.07	**0.28±0.01**	**0.02±0.00**	**0.04±0.00**	**0.11±0.02**

Prediction invariance, variance of risk across domains:

Method	Camelyon17	FMoW	PACS	TerraIncognita	DomainNet
ERM	1.55±0.27	139±50.0	9.89±2.09	12.7±2.6	553±26.5
GroupDRO	0.93±0.11	161±171	398±65.1	603±22.3	668±17.9
RSC	1.90±0.50	181±70.8	6.34±0.91	10.3±3.6	631±7.06
MixStyle	1.67±0.91	110±88.9	6.51±1.20	9.10±0.56	563±4.68
DSU	3.85±1.30	237±136	16.2±5.24	10.6±1.5	567±21.1
LISA	1.81±1.14	111±15.2	3.02±0.47	9.39±0.51	520±11.9
Fish	5.68±1.81	251±45.3	9.08±5.38	9.37±1.59	567±13.2
XDomainMix	**0.90±0.28**	**109±11.7**	**2.10±0.21**	**8.22±1.05**	**504±15.8**

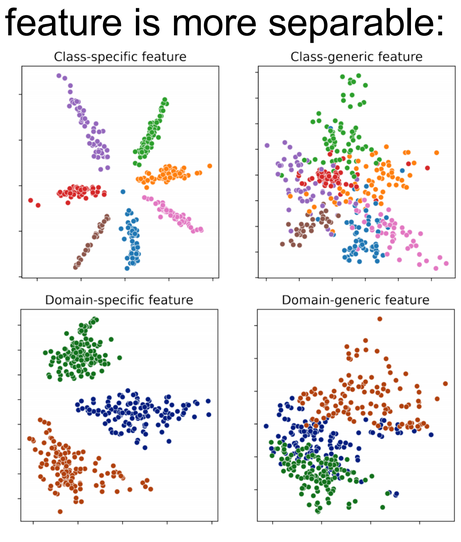
Diversity of Augmented Features

Maximum mean discrepancy to measure the distance between original and augmented features:

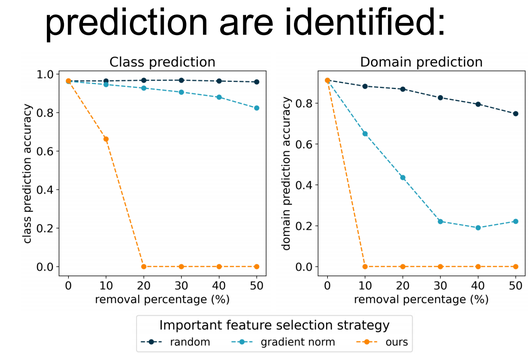
Method	Camelyon17	FMoW	PACS	TerraIncognita	DomainNet
MixStyle	6.65±0.18	8.58±0.50	3.11±0.34	3.20±0.26	2.81±0.11
DSU	26.77±2.53	20.93±1.87	6.97±0.03	10.85±0.32	5.75±0.01
XDomainMix	**38.82±0.28**	**34.06±0.11**	**14.82±0.11**	**14.36±0.12**	**10.27±0.02**

Identified Features

Identified class/domain-specific feature is more separable:



Important feature for prediction are identified:



Ablation

Mixing either $Z_{c,d}$ or $Z_{\neg c,d}$ improves domain generalization.

Discard $Z_{c,d}$ also improves domain generalization performance.

mix $Z_{c,d}$	mix $Z_{\neg c,d}$	discard $Z_{c,d}$	Camelyon17	FMoW
			70.3±6.4	32.3±1.3
✓			78.3±5.5	32.9±2.2
	✓		79.1±6.0	33.6±1.1
✓	✓		79.6±7.0	31.9±0.4
✓	✓	✓	**80.9±3.2**	**35.9±0.8**

link to paper

link to code